LOCUS: AI-Driven Household Context-Awareness for Predictive Cleaning

Team Members

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I. Introduction

Motivation: Why are we doing this?

This is to address current limitations in robotic vacuum cleaners, which have evolved from simple tools to essential smart appliances but remain reactive rather than proactive. Despite hardware advances, current commercial models only operate on preset schedules or direct commands, lacking the ability to respond immediately to dynamic contamination scenarios or understand context such as when, where, and why dirt occurs. Furthermore, privacy concerns arise because conventional systems transmit sensitive data to cloud servers. This approach enables proactive, context-aware cleaning and ensures data privacy by processing sensitive information locally.

What do we want to see at the end?

- Predictive Cleaning Based on Multi-modal Context: Complete an 'Active Partner' system that goes beyond simple scheduling by fusing visual(YOLO), auditory(YAMNet), and positional(Apple ArKit) information to predict the timing and area of contamination and respond accordingly.
- On-device AI-based Privacy Assurance: Process sensitive raw data (video, audio) internally on the edge device without transmitting it to the cloud, and fundamentally prevent privacy leakage by sharing only non-personally identifiable model weights through Federated Learning.
- User-specific Environmental Adaptation: Learn the semantic labels assigned to spaces and the user's living patterns via an app, providing a personalized cleaning solution optimized for each household.

II. Datasets

1. YOLO Datasets (Visual Feature Extraction)

- Objective: Household object recognition and contamination detection for the visual perception of robotic vacuums.
- Final Data: HomeObjects-3K + HD10K

Dataset	Purpose	Total	Train / Val	Source
HomeObjects-3K	Household Object Recognition	2,689	2,285 / 404	Ultralytics, License: AGPL-3.0
HD10K	Contamination Detection	6,000	4,800 / 1,200	IROS 2022 Dataset
Final Data	YOLO Training	8,689	7,085 / 1,604	

- Classes(14): bed, sofa, chair, table, lamp, tv, laptop, wardrobe, window, door, potted_plant, photo_frame, solid_waste, liquid_stain
- YOLO Training Settings:
  - Base Model: YOLOv11n (COCO pretrained)
  - Freeze: 10 layers
  - Learning Rate: 0.001
  - Epochs: 50 (Early Stopping → 24)
  - Image Size: 640×640, Batch Size: 4

2. YAMNet Datasets (Auditory Feature Extraction)

- Objective: Context awareness based on household environmental sounds for robotic vacuums (Auditory).
- Source: Google AudioSet → Refined Home Audio Dataset constructed after background/noise removal.
- Final Data: Total 22,212 clips(Train 17,770 / Val 2,221 / Test 2,221)
  - Classes(17): speech, door, dishes, cutlery, chopping, frying, microwave, blender, water_tap, sink, toilet_flush, telephone, chewing, television, footsteps, vacuum, hair_dryer
  - Input Shape: (N, 1024)
- YAMNet Training Settings:
  - Backbone: YAMNet (MobileNetV1-based, 1024-dim, Frozen)
  - Head: Dense(512-ReLU) → Dropout(0.2) → Dense(17-Sigmoid)
  - Loss: Binary Cross Entropy (Weighted for class imbalance)
  - Epochs: 100 (Early Stopping)
  - Final Model: TFLite FP16, 17MB

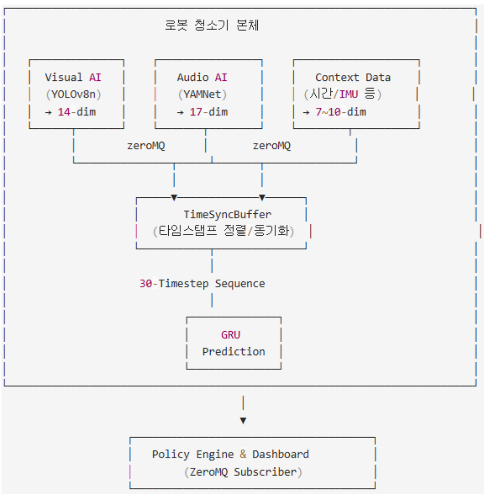
3. GRU Datasets (Context-Based Cleaning Prediction)

- Objective: Prediction of zones requiring cleaning based on context.
- Input: YOLO(14) + YAMNet(17) + Pose(51) + GPS(4) + Time(10) → 160-dimensional context vector generated by AttentionContextEncoder.
- Sequence Length: 30 (Each timestep = 1 context vector)
- Output: Probability of cleaning necessity for 4 zones (balcony, bedroom, kitchen, living_room), Multi-label binary classification.
- Data Count: Train 1,120 / Val 280

III. Methodology

1. Choice of Algorithms & Models

1. **Visual Context: YOLOv11n-pose (Quantized)**
    - We selected YOLOv11n, the most lightweight version of the YOLO model family. Specifically, we adopted the Pose version to go beyond simple object detection and utilize Keypoints to precisely analyze human actions (e.g., 'sitting', 'standing').
    - Optimization: We applied TensorFlow Lite Quantization for deployment on edge device environments. This effectively reduced the model size while maintaining inference speed.
2. **Audio Context (Transfer learning, Quantized)**
    - YAMNet is designed based on the MobileNetV1 architecture, which is optimized for mobile devices. It offers the advantage of efficiently classifying household environmental sounds (such as dishwashing) without the high computational costs associated with large audio transformer models.
3. **Sequential Prediction: GRU (Gated Recurrent Unit)**
    - We selected GRU over LSTM for time-series prediction. GRU has a simpler gate structure than LSTM, requiring less computation, making it more suitable for edge devices. Simultaneously, it effectively captures the Temporal Dependencies of household living patterns.
4. **Data Synchronization: TimeSyncBuffer (Custom Algorithm)**
    - Standard synchronization is difficult because vision sensors (~30fps) and audio sensors (~1s) operate at different frequencies. Therefore, we implemented a custom Sliding Window algorithm that aligns asynchronous data streams based on timestamps (using the LOCF strategy) to merge them into a single context vector.
5. **Federated Learning: FedPer Architecture**
    - We adopted the FedPer architecture to balance privacy protection and model performance. Unlike standard FedAvg, this approach separates the model into a Base Layer (Shared) and a Head Layer (Personal), allowing the robot to learn unique household habits without sharing sensitive label distribution information.



IV. Evaluation & Analysis

1. YOLO Object Detection Model Results

- Validation Metrics

Metric	Score
Precision	0.637
Recall	0.544
mAP50	0.566
mAP50-95	0.389

- Loss Values

Loss	Values
Box Loss	0.988
Class Loss	0.793
DFL Loss	0.954

⇒ Visual object recognition performance is at a moderate level.

2. YAMNet Model Results

- Overall Performance Summary

Item	Value
Total Samples	22,212
Macro AUC	0.9899
Macro AUPR	0.9705

- Per-Class Performance(Top5)

Rank	Class	AUC	AUPR
1	vacuum	0.9980	0.9957
2	hair_dryer	0.9981	0.9941
3	toilet_flush	0.9962	0.9911
4	blender	0.9948	0.9886
5	telephone	0.9954	0.9870

⇒ Accurately classifies auditory events.

3. AttentionContextEncoder Model Results

- Loss: 0.0065 → 0.0001 (Rapid convergence within 5 Epochs)
- Accuracy: Train ~73%, Val ~68% (Stable)
- AUC: 1.0 (Train/Val) → Clear classification boundaries
- Precision: 1.0 (No False Positives)

⇒ The AttentionContextEncoder stably learns context vectors without overfitting.

4. GRU Model Results

- Training Convergence: Initial Loss 0.28 → Dropped to under 0.05 within 5 Epochs (~94% reduction).
- Final Loss: Train Loss 0.005, Validation Loss 0.001 → Very stable convergence.
- No Overfitting: Validation Loss is consistently lower than Train Loss.
- Accuracy: Balanced maintenance of Train 73.4% and Validation 68.3%.

⇒ The model demonstrates generalization capability without overfitting to the training data, ensuring reliable contamination detection